

CURRICULUM VITAE

Yuval Shmilovitz

INSTAAR Postdoctoral Associate

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EDUCATION

- 2024–present PostDoctoral fellowship at the University of Colorado Boulder, Cooperative Institute for Research in Environmental Sciences (CIRES), the Institute of Arctic and Alpine Research (INSTAAR), and the Community Surface Dynamics Modeling System (CSDMS).
- 2019-2023 Ph.D. (direct track) at The Hebrew University of Jerusalem, the program of Hydrology and Water Resources. Advisors: Prof. Efrat Morin (HUJI), Prof. Yehuda Enzel, (HUJI), Prof. Itai Haviv (BGU). Thesis: *Climatic imprints on arid cliff-related hillslope evolution: linking hydrogeomorphic processes across time scales*
- 2017-2019 M.Sc. at The Hebrew University of Jerusalem, the program of Hydrology and Water Resources. Advisors: Prof. Efrat Morin (HUJI), Prof. Yehuda Enzel, (HUJI), Prof. Itai Haviv (BGU). Thesis: *Linking storm-scale rainfall regime with arid slope morphology and evolution*
- 2014-2017 B.Sc. with high honor (Magna cum laude) at The Hebrew University of Jerusalem, Earth Sciences and History departments

PROFESSIONAL EXPERIENCE, VISITS & TRAINING

- 2026 – present Postdoctoral Associate, Institute of Arctic and Alpine Research (INSTAAR), and the Community Surface Dynamics Modeling System (CSDMS), University of Colorado Boulder, CO, USA
- 2024 – 2026 PostDoctoral fellow, Cooperative Institute for Research in Environmental Sciences (CIRES), University of Colorado Boulder, CO, USA
- May 2022 – July 2022 Visiting scientist, The Community Surface Dynamics Modeling System (CSDMS), University of Colorado Boulder, CO, USA
- Aug. 2020 Earth Surface Processes Institute (ESPI) summer school, University of Colorado Boulder, CO, USA
- Aug. 2018 Runoff Prediction in Ungauged Basins (PUB) summer school, Vienna University of Technology, Vienna, Austria
- June 2018 – Aug. 2018 Research Assistant, excellent student's summer project, The Hebrew University of Jerusalem, Jerusalem, Israel. Supervisor: Prof. Dov Avigad

GRANTS AND FELLOWSHIPS

- 2024-2027 Jewish National Fund (KKL-JNF) Forestry Division (\$120,000)** *Rainstorms and climate change impacts on soil erosion in the Northern Negev and effective preservation strategies*
- 2024-2026 CIRES Postdoctoral Grant, University of Colorado Boulder (\$150,000/\$150,000) (sole PI) *Understanding hydrogeomorphic impacts of rainstorm variability on dryland headwater catchments*
- 2022 Jewish National Fund (KKL-JNF) fellowship for climate change studies (\$13,500/\$13,500) (sole PI) *Soil erosion and characterization of extreme erosive events in arid areas*
- 2021-2025 Collaborative Research, BSF-NSF (EAR-2100702/2100753) (\$239,902/\$239,902)* *Deciphering the role of extreme rainstorms and hydroclimatic regime on arid escarpment retreat and sub-cliff slope evolution*

*Yuval Shmilovitz wrote the research proposal, but is not the official PI

**Yuval Shmilovitz is a research associate and collaborator who played a significant role in developing and writing the research proposal, but is not the official PI

AWARDS AND HONOURS

- 2026 Dick Chorley Award, British Society for Geomorphology
- 2024 Bentor Award (for an excellent geoscience dissertation in Israel), Institute of Earth Sciences, The Hebrew University of Jerusalem
- 2024 CIRES Postdoctoral fellowship, University of Colorado Boulder
- 2023 Peretz Grader award for an excellent early-career scientist, Israel Geological Society
- 2022 Student Research Award for Desert Research (Farouk El-Baz), Geological Society of America
- 2021 Outstanding paper by a PhD student, Israel Geographical Society
- 2018, 2019, 2020 The Advanced School for Environmental Studies' scholarship for academic excellence (awarded 3 times), The Hebrew University of Jerusalem
- 2015 Institute of Earth Sciences summer scholarship for excellent students, The Hebrew University of Jerusalem
- 2014, 2015, 2016 Dean's list for academic excellence (awarded 3 times), The Hebrew University of Jerusalem

TEACHING EXPERIENCE

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|-------------------|------------------------------------|--|
| 2021 – 2023 | The Hebrew University of Jerusalem | Summer Field Camp
(field teaching, B.Sc., 2nd year) |
| 2020, 2021, 2023 | The Hebrew University of Jerusalem | Introduction to Geomorphology
(frontal & field teaching, B.Sc., 2nd/3rd year) |
| 2020 – 2022 | The Hebrew University of Jerusalem | Introduction to Geological Mapping
(field teaching, B.Sc., 2nd year) |
| 2020 | The Hebrew University of Jerusalem | Northern Negev Field Trip
(field teaching, B.Sc., 2nd year) |
| 2019, 2021 – 2023 | The Hebrew University of Jerusalem | Geological Excursion in the Negev
(field teaching, B.Sc., 2nd year) |
| 2017 – 2022 | The Hebrew University of Jerusalem | The Dynamic Earth
(frontal teaching, B.Sc., 1st year) |

OUTREACH, SUPERVISING AND MENTORING

- May 2025 & July 2024 Project mentor, The Earth Surface Processes Institute (ESPI), CSDMS, University of Colorado Boulder
- 2021 - 2023 Research project instructor (youth program), 'Kadim' educational center, Israel, *Geomorphology and hydrology across the Ramon cliffs and slopes*
- 2021 - 2022 Bachelor project mentor, The Hebrew University of Jerusalem, *Optimization of stochastic rainfall generator for arid regions*
- 2020 - 2022 Bachelor project mentor, The Hebrew University of Jerusalem, *Rainfall spatiotemporal analysis in the central Negev using a convection-permitting climate model*

REVIEWER SERVICE

- 2026 *Earth Surface Processes and Landforms* (1)
- 2025 *NSF proposal ad hoc reviewer* (1), *Earth Surface Processes and Landforms* (1), *Journal of Geophysical Research: Earth Surface* (2)
- 2024 *Geophysical Research Letters* (1), *Journal of hydrology* (2)
- 2023 *Water Resources Research* (2)

CONFERENCE SESSIONS CONVENED

- 2026 Co-convener (GM 2.6): Measuring, modelling and forecasting hydrogeomorphic processes across scales (at EGU annual meeting)
- 2024 Co-convener (GM 4.2): Hydrogeomorphic cascading processes and hazards impacted by environmental changes and extreme events (at EGU annual meeting)
- 2023 Lead-convener (GM 11.1): Geomorphic perspectives of environmental changes and extreme events in drylands and beyond (at EGU annual meeting)
- 2022 Lead-convener (GM 10.2/CL 3.1/HS 13/NH 1): From hydro-climatology to hydro-geomorphology under extreme climatic events (at EGU annual meeting)
- 2022 Co-convener: Graduate student conference, Water Research in Israel, Technion, Haifa, Israel
- 2019 Co-convener: The 4th student seminar on innovations in hydrology and water resources, the Hebrew University of Jerusalem

ORGANIZATION OF RESEARCH EXPEDITIONS

- 22/06/2022–25/06/2022 Book Cliffs (Utah, USA).
Activities: topographic and vegetation survey. Cosmogenic nuclides sampling
- 02/06/2024–07/06/2024 Book Cliffs (Utah, USA).
Activities: topographic survey, hydrological measurements, soil sampling

INVITED TALKS

- Dec. 2025 **Shmilovitz, Y.**, Rossi, M. W., Gensini, V., Ashley, W., Haberlie, A., & Tucker, G. E. Climate-induced sediment budget transitions in dryland catchments: insights from landscape evolution numerical experiments across timescales, AGU 2025 annual meeting (withdrawn due to inability to attend)
- Oct. 2025 **Shmilovitz, Y.**, Earthcasting across timescales: modern to Holocene-scale perspective on climate-induced sediment budget transitions, Geology Department Colloquium, University of Colorado Boulder
- May 2025 **Shmilovitz, Y.**, *On rainfall variability, erosion, and evolution of ephemeral landforms*, The Earth Surface Processes Institute, University of Colorado Boulder
- Feb. 2025 **Shmilovitz, Y.**, *Integration of high-resolution climate variability in the quantification of floods, erosion, and landscape evolution of ephemeral landforms*, Southwest Watershed Research Center, Tucson, AZ, USA
- June 2024 **Shmilovitz, Y.**, *On rainfall variability, erosion, and evolution of ephemeral landforms*, Colorado State University
- Nov. 2022 **Shmilovitz, Y.**, *Modeling long-term arid cliffs and sub-cliff slopes evolution under short-duration extreme rainstorm events*, CSDMS fall webinar
- April 2021 **Shmilovitz, Y.**, *Frequency analysis of storm-scale soil erosion and characterization of extreme erosive events by linking process-based erosion model and a stochastic rainfall generator*, Volcani Institute, Israel

PRESS AND QUOTATIONS

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| 2025 | <i>A Disaster foretold: The Failures That Led to the Deadliest Flood of the 21st Century</i> | Yedioth-Net (ynet, in Hebrew) |
| 2025 | <i>Dams failures intensified 2023 flood tragedy in Libya</i> | CIRES Newsroom |
| 2024 | <i>Rainstorm Intensity Drives Desert Landscape Evolution</i> | AGU's EOS (Editor's highlights) |
| 2023 | <i>Desert Landscape Evolution Controlled by Storm Intensity</i> | AGU's EOS (Editor's highlights) |
| 2021 | <i>Researchers are using ICESat-2 elevation data to create bathymetry maps of shallow lakes in the remote desert</i> | NASA Earth Observatory |

PUBLICATIONS

- 12 Wei, H., Polyakov, V., Goodrich, D., Morin, E., Guertin, P. D., Assouline, S., Heilman P., Unkrich, C., **Shmilovitz, Y.** Marra, F., (2025). Modeling Runoff and Sediment Yield at the Event scale in Semiarid Watersheds. *International Soil and Water Conservation Research*. <https://doi.org/10.1016/j.iswcr.2025.07.001>.
- 11 Armon, M., **Shmilovitz, Y (equal contribution)**., & Dente, E. (2025). Anatomy of a foreseeable disaster: Lessons from the 2023 dam-breaching flood in Derna, Libya, *Science Advances*, 11(13), eadu2865. <http://doi.org/10.1126/sciadv.adu2865>
- 10 **Shmilovitz, Y.**, Rossi, M. W., & Tucker, G. E. (2025). Multi-century erosion and landscape evolution of ephemeral catchments in response to sub-daily rainfall distribution changes, *Geophysical Research Letters*, 52(5), e2024GL113179. <https://doi.org/10.1029/2024GL113179>
- 9 **Shmilovitz, Y.**, Tucker, G. E., Rossi, M. W., Morin, E., Armon, M., Pederson, J., Campforts, B., Haviv, I., & Enzel, Y. (2024). Impacts of rainstorm intensity and temporal pattern on caprock cliff persistence and hillslope morphology in drylands. *Journal of Geophysical Research: Earth Surface*, 129(2), e2023JF007478. <https://doi.org/10.1029/2023JF007478>
- 8 **Shmilovitz, Y.**, Marra, F., Enzel, Y., Morin, E., Armon, M., Matmon, A., Mushkin, A., Levi, Y., Khain, P., Rossi, Matthew W., Tucker, G., Pederson, J., & Haviv, I. (2023). The impact of extreme rainstorms on escarpment morphology in arid areas: Insights from the central Negev Desert. *Journal of Geophysical Research: Earth Surface*, 128(10), e2023JF007093. <https://doi.org/10.1029/2023JF007093>
- 7 **Shmilovitz, Y.**, Marra, F., Wei, H., Argaman, E., Goodrich, D., Assouline, S., Morin, E. (2023). Assessing the controlling factors on watershed soil erosion during intense rainstorm events using radar rainfall and process-based modeling. *Catena*, 231, 107282. <https://doi.org/10.1016/j.catena.2023.107282>
- 6 **Shmilovitz, Y.**, Shelef, E., Wieler, N., Zhang, H., Mushkin, A. (2023). Estimating the age of abandoned alluvial surfaces using morphologic dating of gully incision. *Journal of Geophysical Research: Earth Surface*, 128(3), e2022JF006875. <https://doi.org/10.1029/2022JF006875>
- 5 **Shmilovitz, Y.**, Enzel, Y., Morin, E., Armon, M., Matmon, A., Mushkin, A., Pederson, J., Haviv, I. (2023). Aspect-dependent bedrock weathering, cliff retreat, and cliff morphology in a hyperarid environment. *GSA Bulletin* 2022; 135 (7-8): 1955–1966. <https://doi.org/10.1130/B36442.1>
- 4 Wieler, N., Mushkin, A., Zhang, H., Sagi, A., Porat, N., **Shmilovitz, Y.**, Zhikun, R., Feipeng H., Piong, S., Jinrui, L., Shelef, E. (2022). Geomorphic Dating of Across-Fault Gully Incision Reveals Time-Invariant Late Quaternary Slip-Rates at the Eastern Termination of the Altyn Tagh Fault. *Geophysical Research Letters*, 49(8), e2021GL096933. <https://doi.org/10.1029/2021GL096933>
- 3 **Shmilovitz, Y.**, Marra, F., Wei, H., Argaman, E., Nearing, M., Goodrich, D., Assouline, S., Morin, E. (2021). Frequency analysis of storm-scale soil erosion and characterization of extreme erosive events by linking the DWEPP model and a stochastic rainfall generator. *Science of The Total Environment*, 787, 147609. <https://doi.org/10.1016/j.scitotenv.2021.147609>
- 2 **Shmilovitz, Y.**, Morin, E., Rinat, Y., Haviv, I., Carmi, G., Mushkin, A., Enzel, Y. (2020). Linking frequency of rainstorms, runoff generation and sediment transport across hyperarid talus-pediment slopes. *Earth Surface Processes and Landforms*, 45(7), 1644-1659. <https://doi.org/10.1002/esp.4836>
- 1 Armon, M., Dente, E., **Shmilovitz, Y.**, Mushkin, A., Cohen, T. J., Morin, E., Enzel, Y. (2020). Determining bathymetry of shallow and ephemeral desert lakes using satellite imagery and altimetry. *Geophysical Research Letters*, 47(7), e2020GL087367. <https://doi.org/10.1029/2020GL087367>

SUBMITTED PUBLICATIONS

- 13 **Shmilovitz, Y.**, Rossi, M. W., Gensini, V., Ashley, W., Haberlie, A., & Tucker, G. E. Rainburst tail dominates landscape degradation through the Holocene and into the future. Under review.

CONFERENCE PRESENTATIONS

- 18 **Shmilovitz, Y.**, Tucker, G. E., Morey, M. S., Campforts, B., Gabel, V., and Hutton, E.(2026). Extended Gravel-Bedrock Eroder 1.0: a Landlab component for sediment and bedrock dynamics across a range of river systems, Oral presentation, European Geosciences Union (EGU) 2026 annual meeting.
- 17 **Shmilovitz, Y.**, Rossi, M. W., Gensini, V., Ashley, W., Haberlie, A., & Tucker, G. E. (2025). Eco-hydrology and lithological controls on ephemeral catchments landscape evolution: insights from the High Plains (Colorado) using high-resolution climate projections. Poster, CIRES Rendezvous 2025.
- 16 **Shmilovitz, Y.**, Rossi, M. W., Gensini, V., Ashley, W., Haberlie, A., & Tucker, G. E. (2025). Eco-hydrology and lithological controls on ephemeral catchments landscape evolution: insights from the High Plains (Colorado) using high-resolution climate projections. Poster, CSDMS 2025 annual meeting.
- 15 **Shmilovitz, Y.**, Rossi, M. W., Gensini, V., Ashley, W., Haberlie, A., & Tucker, G. E. (2025). Translating high-resolution climate change projections into erosion-vegetation feedbacks, sediment dynamics, and multi-century topographic evolution of dryland catchments. Oral presentation, EGU annual meeting.
- 14 **Shmilovitz, Y.**, Rossi, M., Tucker, G., (2024). Multi-century Sediment Yield and Topographic Evolution of Gullied Catchments: an Event-based Approach to Quantify the Effects of Changes in Rainfall Distribution, Oral presentation, AGU 2024 annual meeting.
- 13 **Shmilovitz, Y.**, Marra, F., Enzel, Y., Morin, E., Armon, M., Matmon, A., Mushkin, A., Levi, Y., Khain, P., Rossi, Matthew W., Tucker, G., Pederson, J., & Haviv, I. (2023). The impact of extreme rainstorms on escarpment morphology in arid areas: Insights from the central Negev Desert. The Geological Society of America Annual Meeting (GSA Connects 2024). Oral presentation (presented by YE).
- 12 **Shmilovitz, Y.**, Rossi,M., Tucker, G., (2024). Gully development and soil erosion: An event-based approach using Landlab, Poster, CSDMS annual meeting.
- 11 Dente, E., Armon, M., & **Shmilovitz, Y.** (2024). The September 2023 flood in Derna, Libya: an extreme weather event or man-made disaster? Oral presentation, EGU annual meeting (presented by ED).
- 10 **Shmilovitz, Y.**, Rossi, M., Tucker, G., Campforts, B., Pederson, J., Morin, E., Armon, M., Enzel, Y., Haviv, I.(2023). Simulating dryland cliffs evolution in response to extreme rainstorms, Poster, EGU annual meeting.
- 9 **Shmilovitz, Y.**, Shelef, E., Wieler, N., Huiping, Z., Mushkin, A., (2022). Estimating the age of abandoned alluvial surfaces using morphologic dating of gully incision, Oral presentation, GSA Annual Meeting 2022 (presented by AM).
- 8 **Shmilovitz, Y.**, Marra, F., Enzel, Y., Morin, E., Armon, M., Matmon, A., Mushkin, A., Levi, Y., Khain, P., Haviv, I., (2022). The signature of extreme rainstorms properties on cliff morphology in arid areas, Pico, EGU annual meeting.
- 7 **Shmilovitz, Y.**, Marra, F., Wei, H., Argaman, E., Nearing, M., Goodrich, D., Assouline, S., Morin, E., (2021). Frequency analysis of storm-scale soil erosion and characterization of extreme erosive events by linking process-based erosion model and a stochastic rainfall generator, Oral presentation, Eyal annual meeting, Zichron Yaakov, Israel.

- 6 **Shmilovitz, Y.**, Mushkin, A., Wieler, N., Zhang, H., Shelef, E., (2021). Morphologic dating of gully incision on abandoned alluvial surfaces using a landscape evolution model, Oral presentation, The Israel Geological Society Annual Conference, Yerocham, Israel.
- 5 **Shmilovitz, Y.**, Marra, F., Wei, H., Argaman, E., Nearing, M., Goodrich, D., Assouline, S., Morin, E., (2020). The role of rainstorm properties on crop-land soil erosion: coupling event-scale modeling with a stochastic rainfall generator for estimating erosion risks, Poster, AGU annual meeting.
- 4 **Shmilovitz, Y.**, Enzel, Y., Morin, E., Armon, M., Matmon, A., Mushkin A., Haviv, I., (2020). Aspect-dependent escarpment retreat, sediment mobilization and slope evolution under hyper-arid conditions., Poster., GSA annual meeting.
- 3 **Shmilovitz, Y.**, Morin, E., Rinat, Y., Haviv, I., Carmi, G., Mushkin, A., Enzel, Y., (2020). Linking frequency of rainstorms, runoff generation and sediment transport across hyperarid talus-pediment slopes, Pico, EGU annual meeting.
- 2 **Shmilovitz, Y.**, Morin, E., Rinat, Y., Haviv, I., Carmi, G., Mushkin, A., Enzel, Y., (2019). Linking frequency of rainstorms, runoff generation, and sediment transport: Implications for arid hillslopes evolution and reorganization, Poster, EGU Leonardo Conference, Esch-sur-Alzetter, Luxembourg.
- 1 **Shmilovitz, Y.**, Morin, E., Rinat, Y., Haviv, I., Carmi, G., Mushkin, A., Enzel, Y., (2019). Linking hydro-metrology and geomorphology across hyper-arid talus-pediment slopes: frequency of runoff generating storms, sheetwash efficiency and form-processes relationships, Oral presentation, The Israel Geological Society annual conference, Kfar-Blum, Israel.